

FRANCISZEK FRASZCZYK

MEng Mechatronics Engineering · Dublin City University
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EDUCATION

MEng Mechatronics
Dublin City University
2021 – 2026

BEng Mechatronics
Dublin City University
2021 – 2026

Leaving Certificate
Cross & Passion College
542 CAO Points · 2021

TECHNICAL SKILLS

- PLC & Automation**
- PLC Programming
 - SCADA Screen Design
 - HMI Networking
 - Commissioning
- AI & Embedded Systems**
- TensorFlow Lite Micro
 - Edge AI
 - Convolutional Neural Networks (CNN)
 - ESP32 Microcontroller
 - Arduino Microcontroller
 - MQTT Communication

- Programming**
- Python
 - C / C++
 - MATLAB
 - OOP

- Engineering**
- Technical Drawings
 - Circuit Analysis
 - Software Development
 - Data Analytics
 - System Simulation & Control
 - Sustainability

- Word & Data Processing**
- Microsoft Office Suite
 - Google Workspace

PROFILE

Mechatronics engineer with a BEng and MEng from DCU, specialising in automation, embedded systems, and AI-driven industrial solutions. Hands-on experience commissioning live industrial plants, programming PLCs, and deploying machine learning models on resource-constrained hardware. Proven ability to bridge software and hardware challenges in real-world manufacturing environments.

EXPERIENCE

- Automation Intern** [GEA Group](#) Mar 2024 – Dec 2024
- Commissioned systems on a live industrial plant, gaining direct exposure to safety-critical operational environments.
 - Programmed and ran PLC code to control industrial processes, from initial setup to full operational capacity.
 - Designed and implemented SCADA screens for operator user interfaces, improving process visibility on the plant floor.
 - Connected and configured HMIs to the plant network, ensuring reliable communication across control systems.
 - Conducted pressure testing of pipework and upheld site health & safety protocols throughout the placement.
 - Worked across multiple machine types, building broad practical knowledge of industrial automation systems.

MENG THESIS

- Predictive Maintenance Using Physical AI** [DCU - Supervisor: Dr. Derek Molloy](#)
- Designed and built a complete end-to-end predictive maintenance (PdM) system for industrial machinery, combining embedded hardware with deep learning inference on the edge.
- Developed vibration-based anomaly detection using a triaxial accelerometer mounted on an experimental motor rig.
 - Iteratively designed and trained a 1D Convolutional Neural Network (1D-CNN), progressing from a regression baseline to a quantised deep learning model.
 - Created physically inspired anomaly augmentation techniques to generate bearing-fault signatures in the absence of labelled fault data.
 - Deployed the quantised model on an ESP32-S3 microcontroller using TensorFlow Lite Micro, achieving real-time inference with minimal memory footprint.
 - Implemented MQTT-based remote monitoring for lightweight transmission of anomaly alerts across the network.
 - Demonstrated robust detection of mechanical resonance and electrically induced imbalance faults across multiple operating speeds.

LANGUAGES

- English – Native
- Polish – Native

CERTIFICATIONS

- SOLAS Safe Pass
- RSA EU Driver License
- CSWA
- Food & Beverage Safety

LinkedIn:



ADDITIONAL WORK EXPERIENCE

Retail Assistant [Circle K](#)

Mar 2025 – Sep 2025

- Customer facing role in a high-volume forecourt environment.
- Operated tills whilst ensuring transactions, log balances and adequate float is maintained.
- Ensured optimal stock rotation to minimise waste.
- Responsible for maintenance of shop floor.

Bartender [Killashee Hotel & Spa, FBD Hotels](#)

2022 – 2024

- Bartender across weddings, restaurant, bistro and function events.
- Trained new staff members and assisted with event setup, reservation management, and bar operations.
- Stock clerk responsibilities including ordering, stock rotation management, and restocking bars.
- Resolved escalated customer queries in a timely and respectful manner.

INTEREST & ACTIVITIES

- Trinity College Dublin Walton Club alumnus.
- Personal Arduino electronics projects.
- Programming.
- Fitness (Fencing, Hiking, Swimming).